

CUSTOMER BEHAVIOR ANALYSIS

Professional Data Analyst Portfolio Report

Python • Jupyter Notebook • SQL • MySQL • Power BI

Customer Shopping Behavior Dataset

End-to-End Data Analyst Workflow

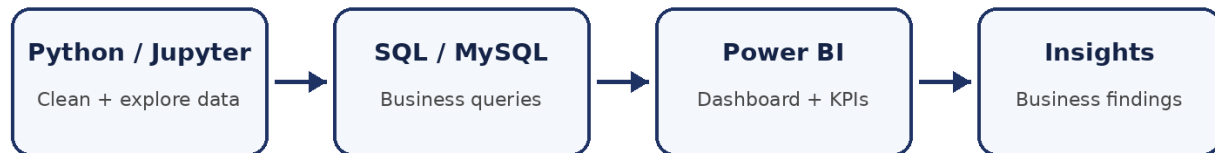


Figure 1. End-to-end workflow used in the project

Portfolio positioning

This project demonstrates a complete data analyst workflow: preparing raw customer data, performing business analysis with SQL, and communicating results through an interactive Power BI dashboard.

1. Executive Summary

The Customer Behavior Analysis project was developed to understand customer purchasing patterns, product performance, subscription behavior, age-group revenue and customer review information. The analysis combines Python in Jupyter Notebook, SQL in MySQL and Power BI.

Python was used as the data preparation and exploratory analysis stage. The notebook checks the dataset, identifies missing values, cleans the review-rating field, standardizes column names and creates useful analysis fields. The prepared dataset was then loaded into MySQL, where SQL queries were used to answer business questions. Power BI was used to present the results as an interactive dashboard.

Project highlights

- Approximately 3.9K customers are represented in the dashboard.
- Displayed average purchase amount: \$59.76.
- Displayed average review rating: 3.75.
- Dashboard includes subscription, gender, category and shipping-type filters.
- Category and age-group analysis is presented through revenue and sales visuals.
- The SQL analysis contains 10 business questions covering customers, products, discounts, shipping, subscriptions and age groups.

2. Dataset & Business Objectives

The source data contains customer-level shopping information such as customer ID, age, gender, item purchased, category, purchase amount, location, season, review rating, subscription status, shipping type, discount usage, previous purchases, payment method and purchase frequency.

Business questions

- How much revenue is generated by male and female customers?
- Which customers used discounts but still spent above the average purchase amount?
- Which products have the highest average review ratings?
- How do Standard and Express shipping customers differ in average purchase amount?
- Do subscribed customers spend more than non-subscribers?
- Which products have the highest discount usage rate?
- How can customers be grouped into New, Returning and Loyal segments?
- What are the top 3 most purchased products in each category?
- Are repeat buyers more likely to subscribe?
- What is the revenue contribution of each age group?

These questions connect the technical work to practical business decisions such as marketing, customer retention, product planning and revenue analysis.

3. Python / Jupyter Notebook — Data Preparation

Python was the first analytical stage. The notebook was used to understand the raw dataset before it was used for SQL and Power BI. This is important because clean and consistent data improves the reliability of later analysis.

Step 1 — Load and inspect the dataset

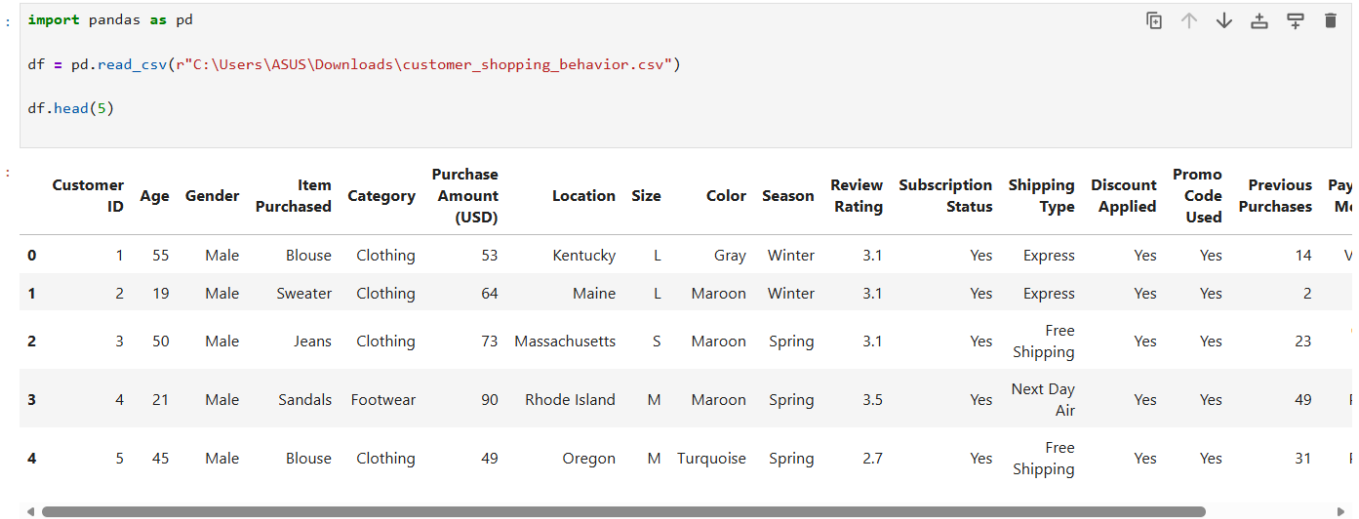


Figure 2. Loading the customer shopping dataset and inspecting the first records

The dataset was loaded with pandas and the first records were reviewed. This helps confirm that the file was read correctly and provides an initial understanding of the available fields.

Step 2 — Check and fix missing values



Figure 3. Missing-value check and category-wise median imputation

The notebook found 37 missing values in Review Rating. Instead of deleting those records, the missing ratings were filled using the median rating within each product category. A second missing-value check confirmed that the review-rating field no longer contained missing values.

Step 3 — Standardize fields and create analysis features

```
35]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')

39]: df.columns

39]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
          'purchase_amount', 'location', 'size', 'color', 'season',
          'review_rating', 'subscription_status', 'shipping_type',
          'discount_applied', 'promo_code_used', 'previous_purchases',
          'payment_method', 'frequency_of_purchases'],
          dtype='str')
```

Figure 4. Column standardization, age-group creation and purchase-frequency conversion

Column names were converted to a consistent lowercase/underscore format. An age_group field was created using age quartiles, and purchase frequency labels were converted into day-based values. These fields make later SQL analysis and Power BI reporting easier.

Important portfolio note

The notebook also connects the prepared dataframe to MySQL and loads it into the customer table. Do not include database passwords or credentials in a public portfolio.

```
from sqlalchemy import create_engine

username = "root"
password = "mahesh"
host = "localhost"
port = "3306"
database = "customer_behaviour"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
)

table_name = "customer"

df.to_sql(
    table_name,
    engine,
    if_exists="replace",
    index=False
)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'")

Data successfully loaded into table 'customer' in database 'customer_behaviour'
```

4. SQL / MySQL — Business Analysis

Once the prepared data was loaded into MySQL, SQL was used to answer business questions. The queries use practical analyst techniques including SELECT, WHERE, GROUP BY, aggregate functions, CASE expressions, common table expressions and window functions.

Query 1 — Subscription and spending analysis

SQL — Customer subscription and spending analysis

```
SELECT subscription_status,  
       COUNT(customer_id) AS total_customer,  
       ROUND(AVG(purchase_amount),2) AS avg_spend,  
       ROUND(SUM(purchase_amount),2) AS total_revenue  
FROM customer  
GROUP BY subscription_status  
ORDER BY avg_spend, total_revenue DESC;
```

Output / result

Purpose:
Compare subscriber vs. non-subscriber customer count,
average spending and total revenue.

Figure 5. SQL query comparing subscribers and non-subscribers

This query compares customer count, average purchase amount and total revenue by subscription status. It helps evaluate whether subscription customers show stronger purchasing behavior.

Query 2 — Revenue contribution by age group

SQL — Revenue by age group

```
SELECT age_group,  
       SUM(purchase_amount) AS total_revenue  
FROM customer  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

Output / result

Purpose:
Measure revenue contribution from each customer age group.

Figure 6. SQL query for age-group revenue contribution

The query groups customers by the age_group created during the Python stage and calculates total revenue. This connects Python feature engineering with SQL business analysis.

Query 3 — Top products within each category

SQL — Top products within each category

```
WITH item_counts AS (  
  SELECT category, item_purchased,  
         COUNT(customer_id) AS total_orders,  
         ROW_NUMBER() OVER (  
           PARTITION BY category  
           ORDER BY COUNT(customer_id) DESC  
         ) AS item_rank  
  FROM customer  
  GROUP BY category, item_purchased  
)  
SELECT item_rank, category, item_purchased, total_orders  
FROM item_counts  
WHERE item_rank <= 3;
```

Output / result

Purpose:
Find the top 3 most purchased products
within every product category.

Figure 7. SQL CTE and ROW_NUMBER analysis for category-level product ranking

A common table expression and ROW_NUMBER() are used to rank products separately within each category. This is a strong example of SQL being used for real business analysis rather than only basic filtering.

Other SQL analyses completed

- Gender-wise revenue comparison.
- Customers who used a discount but spent at or above the average purchase amount.
- Top products by average review rating.
- Average purchase amount by Standard and Express shipping.
- Top products by discount usage rate.
- Customer segmentation into New, Returning and Loyal.
- Repeat-buyer subscription comparison.

The complete SQL script should be kept as a separate portfolio file so recruiters can review the full set of queries.

5. Power BI — Interactive Dashboard

Power BI was used as the reporting and visualization layer. The dashboard brings the customer-level analysis into a single interactive page so a business user can quickly review KPIs and filter the data.

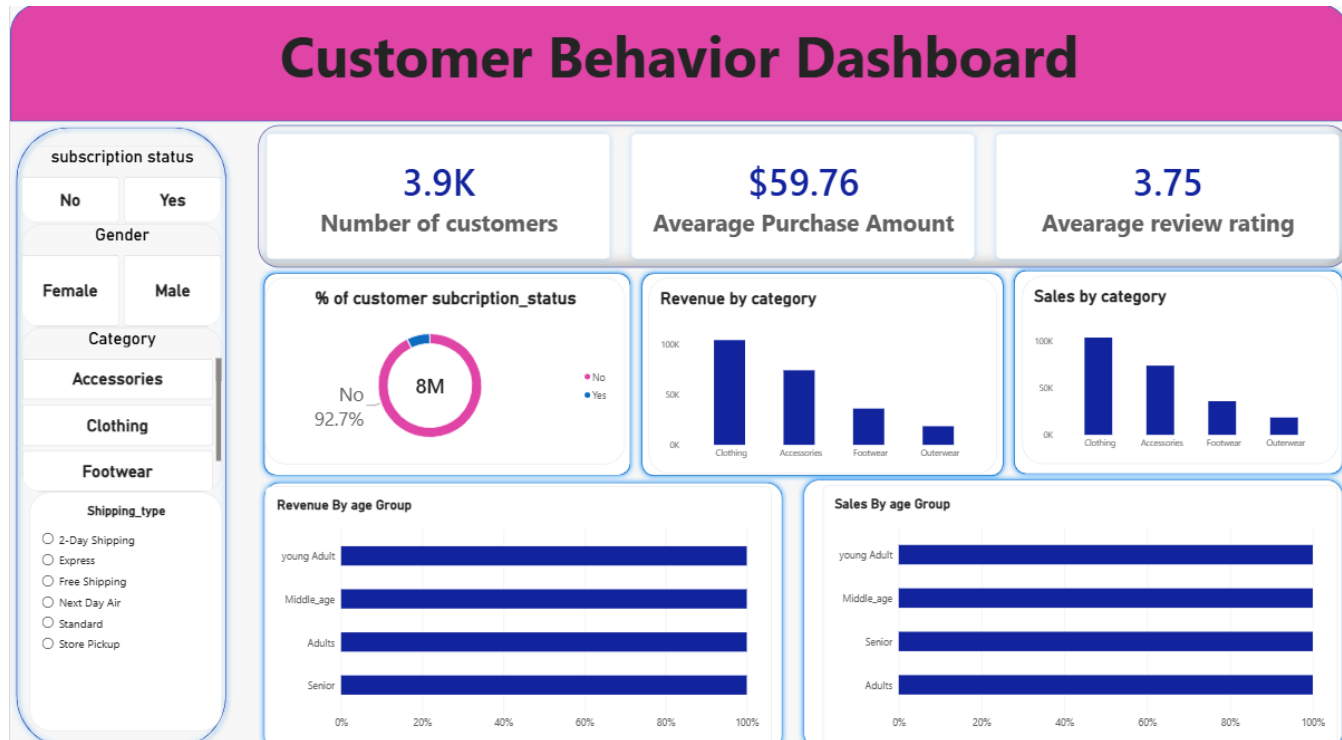


Figure 8. Customer Behavior Dashboard

Dashboard components

- KPI: Number of customers
- KPI: Average purchase amount
- KPI: Average review rating
- Customer subscription-status distribution
- Revenue by category
- Sales by category
- Revenue by age group
- Sales by age group
- Interactive slicers for subscription status, gender, category and shipping type

The dashboard is designed to support quick comparison and interactive exploration. For a portfolio, the .pbix file should be included separately so the recruiter can see the model, measures and interactive functionality.

6. Key Findings & Business Interpretation

Overall customer metrics

- The dashboard displays approximately 3.9K customers.
- The displayed average purchase amount is \$59.76.
- The displayed average review rating is 3.75.

Product and revenue performance

- Clothing is the strongest category in the displayed revenue and sales charts.
- Accessories are the next strongest displayed category, followed by Footwear and Outerwear.
- Category-level revenue and sales views can support inventory and marketing prioritization.

Customer behavior

- The dashboard shows that non-subscribed customers form the majority of the displayed customer base.
- Age-group and subscription filters allow the user to explore customer behavior from different perspectives.
- Shipping type, gender and category filters provide additional segmentation for analysis.

Data-quality / visualization check

Before publishing the dashboard as a final portfolio piece, validate that each age-group chart uses the intended revenue or sales measure and that the subscription donut's center value represents the intended KPI. The report should only publish figures that match the underlying dataset.

7. Business Recommendations

- Focus promotional and inventory planning on high-performing categories such as Clothing.
- Use subscription analysis to design campaigns that convert non-subscribed customers.
- Use age-group revenue analysis to create targeted marketing offers.
- Monitor review ratings and product-level performance to identify products needing improvement.
- Use discount-rate analysis to understand which products depend more heavily on promotions.
- Use repeat-buyer and customer-segment analysis to support retention strategies.

8. Skills Demonstrated

Skill area	Evidence in project	Portfolio value
Python / Pandas	Data loading, missing-value handling, feature engineering	Shows practical data preparation
Exploratory Data Analysis	Dataset inspection and descriptive checks	Shows ability to understand raw data
SQL	10 business questions, aggregations, CTE, CASE, ROW_NUMBER	Shows business-focused querying
Power BI	KPIs, charts, slicers and dashboard design	Shows reporting and visualization
Business Analysis	Insights and recommendations	Shows ability to translate data into decisions

9. Portfolio Deliverables

File	Purpose	Recommended use
Customer_Shopping_Behaviour_Analysis.html / Jupyter Notebook	Python work and outputs	Attach or link as technical evidence
customer_behaviour_dashboard_sql_queries.sql	Complete SQL business queries	Upload with the project
Power BI .pbix file	Interactive dashboard	Upload or share as the main visualization
Customer Behavior Analysis Report.docx	Project explanation and evidence	Use as the professional project report

10. Conclusion

This project demonstrates an end-to-end data analytics process. Python and Jupyter Notebook were used to prepare and explore the customer data, SQL was used to answer business questions, and Power BI was used to communicate the results through an interactive dashboard. The combination of technical analysis and business interpretation makes this project suitable as a Data Analyst portfolio project.